Accelerated Math II/III
HW #9 – End Behavior

Name: _____
Period: _____

Analyzing Graphs of Polynomial Functions

Direction: All problems are to be done on graph paper.

For each function below:

(a) Identify the degree, leading coefficient, end behavior and what happens to $f(x)$ as $x \to \pm\infty$.
(b) Write the polynomial in factored form if it isn't already.
(c) Find the x-intercepts and y-intercepts.
(d) Identify the multiplicity behavior.
(e) Find and plot points between and beyond the x-intercepts.
(f) Draw the graph so that it passes through the plotted points and has appropriate end behavior and multiplicity.

1. $f(x) = -2x^4 + 2x^2$

2. $f(x) = 3x^3 - 15x^2 + 18x$

3. $f(x) = -5x^2 - x^3$

4. $f(x) = -\frac{1}{4}(x-2)^2(x+2)^2$

5. $f(x) = \frac{1}{12}(x+4)(x-3)(x+1)^2$

6. $f(x) = \frac{1}{5}(x-3)^2(x+1)^2$

7. Completely factor $f(x) = x^3 - 2x^2 - 9x + 18$ given that $f(2) = 0$, then find all the zeros and graph the polynomial identifying all parts above